

Supplementary materials: CompMVR: Compression-Aware Multi-View Restoration Using Diffusion Models for Geometrically Consistent 3D Reconstruction

1 Implementation Details

In the first stage, we learn a compression-aware latent representation using paired compressed and uncompressed images. The network is trained with a combination of image reconstruction, QP regression, and codec classification losses, with weights set to 1.0, 0.1, and 0.05, respectively. We use the Adam optimizer with a learning rate of 2×10^{-5} and a batch size of 8. Training images are randomly cropped to 256×256 . We train the first stage for 200K iterations with a single NVIDIA A6000 GPU.

In the second stage, we train the multi-view diffusion restoration model using four input views per sample at a resolution of 512×512 . The model is optimized using AdamW with a learning rate of 5×10^{-5} and a batch size of 2. We adopt LoRA with rank 16, applied to 16 3D attention layers, for efficient fine-tuning. The estimated artifact map $\hat{r}$ is concatenated with the latent representation z_L and projected to an 8-channel feature map before being fed into the UNet. Specifically, the projection fusion $\phi(\cdot)$ (Eq. 4) consists of a single 1×1 convolutional layer that matches the channel dimension of $\hat{r}$ to that of z_L prior to concatenation. We stop gradient backpropagation through $\hat{r}$ to prevent the restoration network from influencing the artifact estimator.

We employ an asymmetric loss for artifact estimation with weight $\lambda_{cae} = 0.3$ and a correspondence-based loss for cross-view consistency with $\lambda_{corr} = 0.01$. To compute the confidence weight w_{pq} used in this correspondence loss (Eq. 8), we leverage VGGT to establish geometric correspondences between view pairs. All view images are resized to 518×518 via bilinear interpolation before being fed into VGGT, we used the GT high-quality images rather than the restored outputs as input. For each view pair, pixel correspondences are obtained by 3D nearest-neighbor matching on the world point maps predicted by VGGT, and the final valid correspondence mask is constructed by retaining only matches that jointly satisfy: (1) a confidence threshold, i.e., predicted confidence exceeding $0.75 \times$ the per-view average confidence; (2) bidirectional mutual nearest-neighbor cycle consistency between the two views ($i \rightarrow j$ and $j \rightarrow i$); and (3) a geometric outlier filter, matching distance below $1.5 \times$ the average matching distance. This stage is trained for 100K iterations on a single NVIDIA A6000 GPU.

2 Quantitative Validation of Compression-Aware Latent Representation

We quantitatively evaluate the Stage 1 latent representation to verify that it captures both codec-aware and continuous compression characteristics. As summarized in Tab. 1, the model achieves 93.17% codec classification accuracy and a QP prediction error (MAE) of 2.93 on unseen QP levels. These results indicate that the learned latent encodes meaningful compression information beyond simple reconstruction. Fig. 1 further analyzes the QP prediction error

across different unseen QP values. The error remains nearly constant across QP intervals, demonstrating that the model generalizes well to compression levels that were not observed during training. This suggests that the latent representation captures continuous compression characteristics rather than overfitting to discrete QP values. Noticeably, the prediction error decreases at higher QP values, likely because stronger compression introduces more distinctive and structured artifacts, which makes the underlying compression level easier to estimate. Overall, these results support that the Stage 1 latent provides a reliable and generalizable compression-aware representation, which serves as an effective conditioning signal for the subsequent multi-view restoration model.

Table 1. Stage 1 evaluation on unseen QP levels. The model achieves high codec classification accuracy and low QP prediction error, demonstrating that the learned latent effectively encodes both codec-specific and continuous compression information.

	Codec Acc (%)	QP MAE
Overall	93.17	2.93
AV1	97.19	3.33
AVC	98.06	2.75
HEVC	80.77	2.17
VP9	96.68	3.47

3 Compression Protocols

Overall setting. We generate compressed inputs from raw YUV420p sequences using FFmpeg. For all codecs, we adopt a random-access

Table 2. VGGT point map estimation results under different codec and QP settings. Overall Chamfer distance is reported, where lower values indicate better point-level geometric accuracy. The **Best** and **Second-best** results are highlighted, respectively.

Codec	QP	Overall ↓				
		Compressed	CODiff*	DiQP*	SIR-Diff*	Ours
AVC	27	0.6447	0.6375	0.6462	0.5891	0.6314
	42	0.6598	0.6597	0.6697	0.6093	0.6591
HEVC	27	0.6382	0.6352	0.6485	0.5997	0.6375
	42	0.6443	0.6481	0.6631	0.6172	0.6407
VP9	27	0.6471	0.6391	0.6406	0.5942	0.6389
	42	0.6401	0.6449	0.6572	0.5978	0.6327
AV1	27	0.6381	0.6371	0.6418	0.5924	0.6364
	42	0.6500	0.6502	0.6526	0.5868	0.6438

* denotes models trained by us on our training datasets.

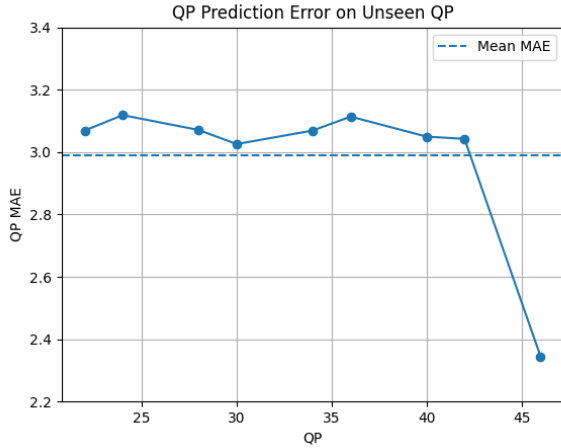


Fig. 1. QP prediction error (MAE) across unseen QP levels. The error remains nearly constant across QP values, indicating that the learned latent captures continuous compression characteristics and generalizes beyond discrete training QP.

(RA) encoding setting with a fixed GOP size of 32 and disabled scene-cut detection, so that the temporal structure is consistently controlled across sequences. For codecs that support B-frame coding, we use three B-frames. After encoding, each bitstream is decoded back to YUV420p format and used as the compressed input for downstream restoration and 3D reconstruction experiments.

AVC. For AVC compression, we use the `libx264` encoder. The target QP value is directly passed to the encoder, and the same fixed GOP configuration is used across all tested QP levels. This allows us to evaluate the effect of AVC compression severity while keeping the temporal coding structure fixed.

HEVC. For HEVC compression, we use the `libx265` encoder. As in AVC, the target QP value is directly used. We additionally constrain the encoder to follow a fixed RA structure by disabling scene-cut adaptation and using a closed-GOP configuration. This avoids unintended changes in the temporal prediction structure across scenes or QP levels.

VP9. For VP9 compression, we use the `libvpx-vp9` encoder. Since VP9 does not use B-frame coding in the same manner as AVC/HEVC, we control the temporal structure mainly through a fixed keyframe interval. The compression strength is controlled using the VP9 CRF parameter, which is obtained by mapping the input QP value to a VP9-specific quality range.

AV1. For AV1 compression, we use the `libsvtav1` encoder. We use the same fixed GOP setting as the other codecs and control the compression strength using the AV1 quantizer parameter. Since the AV1 quantizer is not directly equivalent to AVC/HEVC QP, we use a mapped quality parameter for AV1.

Quantization parameter matching. The quality parameters used by different codecs are not directly equivalent. Therefore, AVC and

HEVC use the specified QP values directly, while AV1 and VP9 use codec-specific quality parameters obtained by a linear mapping from the AVC/HEVC QP range. Specifically, the input QP range is mapped to the AV1 quantizer range and the VP9 CRF range, respectively. In all cases, larger values correspond to stronger compression and lower visual quality. This mapping is intended to provide comparable compression-severity levels across codecs, rather than to make the internal quantization processes identical.

4 3D Evaluation Protocols

3D Gaussian Splatting. For 3D Gaussian Splatting evaluation, we select informative multi-view frames using VGGT-based feature matching. Taking the first frame as the reference view, we rank the remaining frames according to their view-matching relevance and select the top 35 frames with the highest correspondence scores. This selection ensures sufficient view overlap and reliable geometric cues for 3DGS training. We then train and evaluate 3D Gaussian Splatting using the selected train/test views, which are kept identical across all methods and codec settings for a fair comparison.

Feature Matching. We evaluate two-view feature matching on the RealEstate10K dataset to measure whether restored images preserve geometrically reliable correspondences. For each image pair, we use externally extracted correspondences and evaluate them with the ground truth camera intrinsics and relative pose. Following the standard two-view matching protocol, image coordinates and camera intrinsics are consistently resized, and the relative pose is estimated from the matched points using essential-matrix estimation with RANSAC. We then compute the rotation and translation errors with respect to the ground-truth relative pose and report pose AUC at multiple thresholds, together with matching-related statistics such as the number of matches and matching confidence. This evaluation reflects the quality of the restored inputs in terms of multi-view geometric consistency, rather than only image-level perceptual fidelity.

Camera Pose Estimation. We evaluate camera pose estimation on the RealEstate10K dataset using VGGT as the pose estimator. For each scene, we sample a fixed number of frames and use the RealEstate10K metadata to obtain ground-truth intrinsics and camera poses. We compute relative pose errors over all image pairs in the sampled sequence and evaluate the predicted poses using rotation and translation angular errors. The results are reported as relative rotation accuracy (RRA), relative translation accuracy (RTA), and pose AUC at 15 and 30 degrees, averaged across scenes.

Point Map Estimation. We evaluate point map estimation on the CO3Dv2 dataset using VGGT at Tab. 2. For each sequence, a fixed number of frames are sampled and used to predict 3D point maps. The predicted points are aggregated in a common world coordinate frame and filtered using confidence scores. We then align the predicted point cloud to the ground-truth CO3Dv2 point cloud using Umeyama similarity alignment and compute nearest-neighbor distances. We report accuracy, completeness, and overall Chamfer distance to assess the geometric fidelity of the restored multi-view inputs.

Table 3. Per-region full-reference fidelity on the Free dataset. Regions are defined using clean ground truth and shared across all methods. The **Best** and **Second-best** results among restoration methods are highlighted, respectively. Compressed denotes the unrestored input and is excluded from the ranking.

Region	Metric	Compressed	CODiff*	DiQP*	SIR-Diff*	Ours
Smooth	PSNR $\uparrow$	28.42	27.72	24.38	26.88	27.36
	SSIM $\uparrow$	0.805	0.711	0.674	0.681	0.723
Textured	PSNR $\uparrow$	26.94	24.49	22.41	22.16	24.83
	SSIM $\uparrow$	0.781	0.607	0.603	0.526	0.646
Edge-dominant	PSNR $\uparrow$	25.51	20.69	20.61	18.28	21.57
	SSIM $\uparrow$	0.832	0.542	0.649	0.364	0.616

* denotes that the model has been finetuned on our training datasets.

5 Per-Region Fidelity and Failure Cases

Per-Region Fidelity. We evaluate restoration fidelity separately in smooth, textured, and edge-dominant regions of the Free dataset.

Regions are determined from clean ground-truth images using gradient and high-frequency responses. As shown in Tab. 3, CompMVR achieves the best PSNR and SSIM among restoration methods in textured regions and the best PSNR in edge regions. However, its fidelity gap relative to the compressed input increases in high-frequency regions, indicating that fine details removed by compression cannot always be faithfully recovered.

Failure Cases. Fig. 2 shows representative failure cases under severe compression. CompMVR can replace repetitive textures, thin edges, and small structures with perceptually plausible but ground-truth-inaccurate details. This demonstrates the fidelity-plausibility trade-off of diffusion-based restoration and motivates jointly considering perceptual, full-reference, geometric-consistency, and downstream 3D metrics.

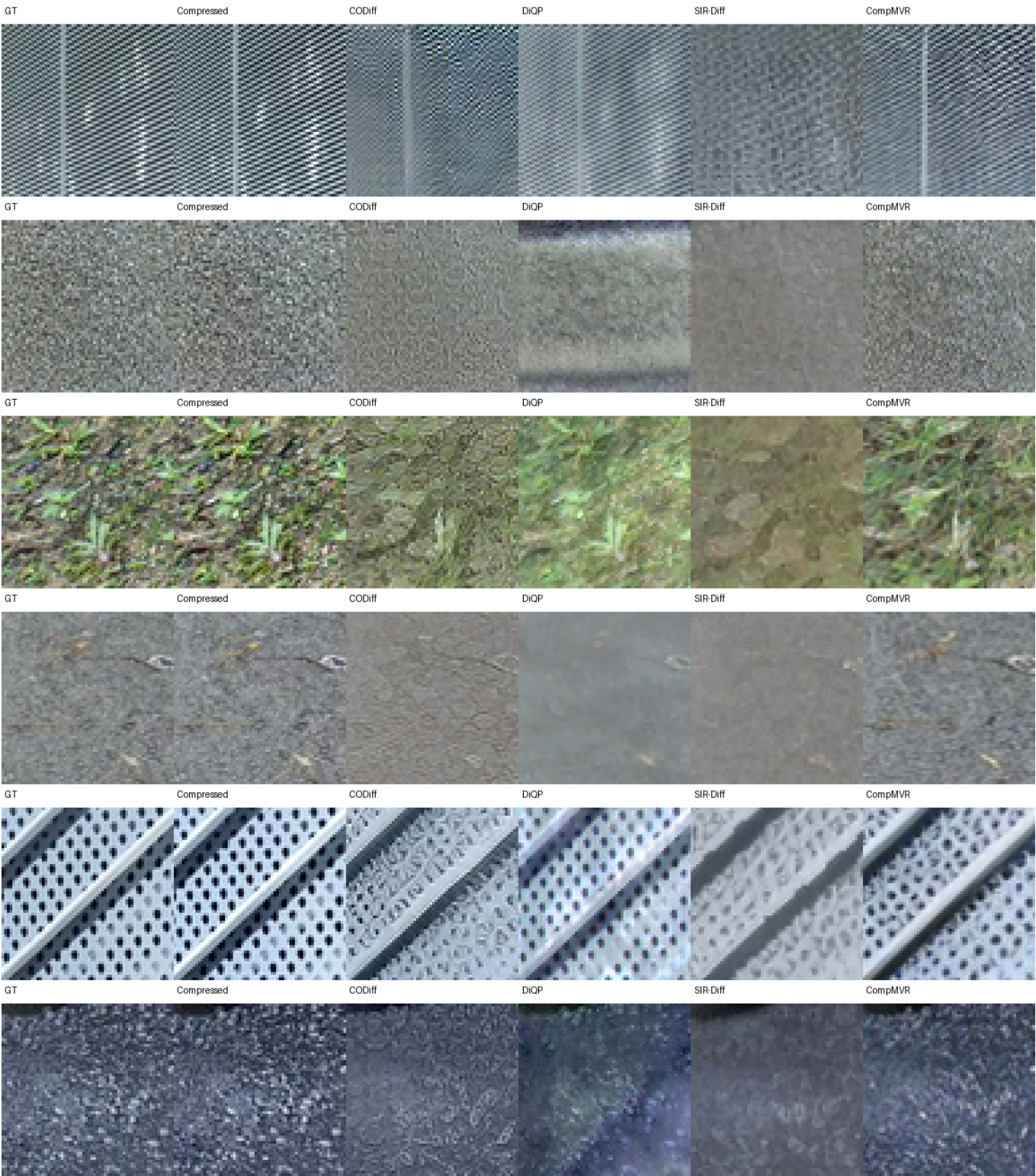


Fig. 2. Representative CompMVR failure cases. Under severe compression, repetitive textures and fine structures can be restored with perceptually plausible but ground-truth-inaccurate details.